Masiko Mamba

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EDUCATION

University of Oklahoma Norman, OK

Master of Science in Computer Science. *Expected May 2027*

Relevant Coursework: Distributed Operating Systems, Computer Security, Database Management, Parallel Networking

University of Oklahoma Norman, OK

Bachelor of Science in Computer Science; Bachelor of Arts in Economics. *May 2026*

Relevant Coursework: Computer Architecture, Machine Learning, Software Quality & Testing, Econometric Analysis

EXPERIENCE

Data Science Intern August 2026 – Present

American Fidelity Assurance *Norman, OK*

- Develop a customer-facing RAG chatbot using Python and PostgreSQL, implementing context-aware retrieval and response generation.
- Build a hybrid retrieval pipeline using BM25, semantic embeddings, and RRF, with deployment in progress.
- Implement conversation-aware retrieval with context windows, semantic similarity, and recency weighting.
- Create a PII-masking and evaluation pipeline to benchmark retrieval and response quality.

Graduate Research Assistant August 2026 – Present

University of Oklahoma, Southern Plains Transportation Center

- Independently architect and deploy a Django/PostgreSQL web platform on Google Cloud with Docker, replacing manual email workflows for approximately 80 users.
- Engineer secure signed-URL uploads to Google Cloud Storage and asynchronous document processing with Cloud Run and Redis.
- Build an Azure OpenAI GPT-4.1 pipeline for compliance checks and summarization, using PyMuPDF to highlight issues directly in PDFs.
- Implement multi-signer PDF approvals, SHA-256 audit trails, resilient reviewer workflows, and automated tests for reliable processing.

Undergraduate Research Assistant May 2025 – May 2026

University of Oklahoma, OUNLP

- Conducted research on improving LLM performance in long-context video understanding and question answering tasks.
- Applied frame and language analysis methods to enhance model accuracy and efficiency in long-context video tasks.
- Evaluated model responses and developed custom evaluation metrics for long-context understanding.

PROJECTS

AI Desktop Assistant

Python, Claude, Qwen, Ollama, Whisper

- Built a Python desktop application that routes natural-language requests between cloud and local LLMs and task-specific actions.
- Implemented an on-device private mode using Qwen and Ollama, keeping conversations and inference local until returning to public mode.
- Integrated Whisper and Qwen-VL for voice commands, screen capture, and contextual analysis of on-screen content.
- Integrated Spotify and Yahoo Finance APIs with file, browser, application, and code-analysis tools to execute user-requested workflows.

SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL

Frameworks & Libraries: Django, React, Node.js, TensorFlow, scikit-learn, numpy, pandas, PyMuPDF

Cloud & Databases: Google Cloud Platform, Cloud Run, Cloud Storage, Cloud SQL, PostgreSQL

AI/ML: RAG, Azure OpenAI, GPT-4.1, Claude, Qwen, Ollama, Whisper, TF-IDF

Tools & Technologies: Git/GitHub, Docker, Linux, REST APIs, Asynchronous Processing, Redis